

Standardized Experimental Framework

Next-Day 24-Hour Hourly Stock Status Forecasting: Task Definition, Dataset Standardization, Metrics & Architectural Protocols

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Executive Objective

This document establishes the single authoritative, standardized experimental specification for all research, dataset processing, model development, metric evaluation, and architectural comparisons across our Next-Day 24-Hour Stock Status Prediction initiatives.

Every experiment in this repository must strictly adhere to the unit of analysis, statistical SKU selection formula, chronological train/validation splitting, loss functions, dynamic thresholding, and objective metric hierarchy defined herein.

0.1 Problem Formulation & Operational Context

0.1.1 Business Context & Research Motivation

In high-frequency fresh retail supply chains (e.g., Dingdong Fresh, instant grocery delivery), stockout events lead to severe operational damage: lost revenue, unfulfilled customer carts, and reduced long-term customer retention.

Traditional inventory forecasting models predict coarse **daily aggregated demand**, which fails to capture intra-day inventory depletion patterns. A store may appear “in-stock” on a daily aggregate basis even if it suffered a critical 4-hour stockout during peak evening sales (5 PM – 9 PM).

To solve this, our framework operates at **hourly granularity**, predicting the 24-hour operational stock status vector for the target day $t + 1$.

0.1.2 Task Definition

Given a sequence of N historical days ($N \in \{7, 10, 14\}$), the objective is to predict the binary stock-status matrix for all active operational hours of the subsequent day ($t + 1$):

$$X_{t-N+1:t} \in \mathbb{R}^{N \times D} \Rightarrow h @ (Y)_{t+1} \in \{0, 1\}^{24}$$

where D represents the input feature dimension, and $y_h = 1$ indicates a stockout state during hour $h \in [1..24]$ while $y_h = 0$ indicates in-stock status.

0.2 Dataset Specification & Preprocessing Standards

0.2.1 Primary Dataset: FreshRetailNet-50K

All standardized experiments utilize the benchmark **FreshRetailNet-50K** dataset (Dingdong Inc.), containing daily and hourly sales, inventory counts, promotion discounts, holiday indicators, and store metadata across thousands of SKUs over multiple months.

0.2.2 Unit of Analysis Definition

Stockout events are localized to store replenishment cycles and regional supply chains. Therefore, the unit of analysis is defined at the strict **store-product time-series level**:

$$\text{series_id} = \text{city_id} + \text{store_id} + \text{product_id}$$

Grouping by product alone or store alone is explicitly prohibited, as inventory levels differ across store locations for the exact same product.

0.2.3 Statistical Top 15 SKU Selection Formula

To eliminate non-moving items and focus evaluation on high-impact, non-trivial demand patterns, SKUs are statistically selected using a weighted multi-factor score:

$$\text{Score} = 0.40 \cdot \text{Sales} + 0.25 \cdot \text{StockoutHours} + 0.20 \cdot \text{StockoutDays} + 0.15 \cdot \text{SalesStd}$$

where all 4 components are min-max normalized across all unique product IDs:

1. **“Sales”**: Total historical sales volume across all stores.
2. **“StockoutHours”**: Cumulative count of historical stockout hours.
3. **“StockoutDays”**: Total days with at least one stockout hour.
4. **“SalesStd”**: Standard deviation of daily sales (measuring demand volatility).

The top 15 products with the highest composite selection scores are retained for standardized benchmark evaluation.

0.2.4 Feature Standardization Protocol

Each time step t is represented by a 10-dimensional standardized feature vector:

1. **sale_amount**: Total daily sales quantity (standardized).
2. **stock_hour6_22_cnt**: Count of operational stockout hours during 6 AM – 10 PM (standardized).
3. **discount**: Average promotional discount rate $[0, 1]$.
4. **holiday_flag**: Binary indicator for national/statutory holidays $\{0, 1\}$.
5. **hours_sale_sum**: Sum of intra-day hourly sales (standardized).
6. **hours_stock_status_sum**: Sum of intra-day hourly stockout status (standardized).
7. **dow_sin**: Cyclical day-of-week sine transformation $\sin(2\pi \cdot \frac{\text{dow}}{7})$.
8. **dow_cos**: Cyclical day-of-week cosine transformation $\cos(2\pi \cdot \frac{\text{dow}}{7})$.
9. **stockout_rolling_3**: 3-day rolling window stockout day count.
10. **sales_momentum**: 1-day sales velocity change $\Delta\text{Sales}_t = \text{Sales}_t - \text{Sales}_{t-1}$.

0.2.5 Chronological Train / Validation Data Splitting

To prevent data leakage and simulate true deployment conditions:

- **Training Set**: Chronological historical data up to cutoff date $T_{\text{train}} = T_{\text{max}} - 15$ days (2.5 months of sequence data).
- **Validation Set**: Final 15 chronological days ($T_{\text{max}} - 15$ days $< t \leq T_{\text{max}}$).
- **Scaler Fitting**: StandardScaler must be fitted **strictly on the training split** and applied to transform both train and validation sets.

0.3 Metric Hierarchy & Objective Measurement Rationale

0.3.1 Primary Operational Metric: Hour-Level F1-Score

In retail datasets, operational hours are dominated by in-stock status (60:40 to 80:20 in-stock ratio). Under such class imbalance:

- Raw Accuracy is **misleading**: A trivial model predicting 100% in-stock achieves 80% accuracy while missing **100% of stockout events**.
- **Hour-Level F1-Score** is our **Primary Objective Metric**:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

It measures the harmonic mean of stockout detection precision and recall without being inflated by true negative in-stock hours.

0.3.2 Secondary Metric Evaluation Protocol

1. **Hour-Level Recall**: Percentage of actual stockout hours correctly detected ($\frac{\text{TP}}{\text{TP} + \text{FN}}$). Crucial for preventing unfilled customer orders.
2. **Exact 24-Hour Match Rate (%)**: Percentage of evaluation days where the model correctly predicts **all 24 hourly stockout statuses perfectly**:

$$\text{ExactMatch} = \frac{1}{M} \sum_{i=1}^M \mathbb{I}(h@(\mathbf{y})_i = \mathbf{y}_i)$$

3. **Mean Absolute Hour Count Error (MAE in hours):** Mean absolute difference between predicted total stockout hours and actual total stockout hours per day:

$$\text{MAE}_{\text{hours}} = \frac{1}{M} \sum_{i=1}^M \left| \sum_{h=1}^{24} h@(\mathbf{y})_{i,h} - \sum_{h=1}^{24} \mathbf{y}_{i,h} \right|$$

4. **Inference Latency (ms/seq):** Average CPU/GPU prediction time per sequence (milliseconds).
5. **Parameter Count:** Total trainable parameters.

0.4 Training Protocol & Optimization Standards

0.4.1 Loss Function Standard: Focal Loss

Standard Binary Cross Entropy (BCE) fails under class imbalance because easy in-stock negative examples dominate the loss gradient. All models are optimized using **Focal Loss** ($\gamma = 2.0, \alpha = 0.5$):

$$\mathcal{L}_{\text{Focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

where $p_t = p$ if $y = 1$ else $1 - p$. This dynamically down-weights easy non-stockout examples and forces the network to learn hard stockout transitions.

0.4.2 Dynamic Threshold Optimization (τ^*)

Fixed classification thresholds ($\tau = 0.50$) are suboptimal under imbalanced Focal Loss outputs. Every model evaluation scans thresholds $\tau \in [0.10, 0.90]$ with step size 0.05 on the validation set to select the optimal threshold τ^* that maximizes the Hour-Level F1-Score:

$$\tau^* = \arg \max_{\tau \in [0.1, 0.9]} F_1(\mathbf{y}, \sigma(h@(\mathbf{z})) \geq \tau)$$

0.5 Standardized Model Architecture Taxonomy

All models evaluated in this repository fall into 5 standardized architectural families:

1. **Baseline Recurrent Models:** Standard LSTMs (hidden sizes $h \in \{16, 32, 64, 96\}$) and Coupled Input-Forget Gate LSTMs (CIFG).
2. **Ablation & Compact Variations:** Window length ablations ($N \in \{7, 10, 14\}$) and feature dimension reductions ($F \in \{4, 6, 10, 13\}$).
3. **Architectural Tweaks:**
 - **Dual-Stream Decoupled Networks:** Parallel Conv1D+GRU streams separating demand and inventory.
 - **Direct Inventory Residual Shortcuts:** Direct linear shortcuts projecting inventory status to output logits.
 - **Dynamic Sigmoid Gated Fusion:** $\alpha = \sigma(W_g[\text{Demand} \parallel \text{Inventory}] + b_g)$.
4. **State-of-the-Art (SOTA) Time-Series Models:**
 - **DLinear:** Trend-Seasonal linear decomposition (6,768 params, 0.3326 ms/seq, F1 = 0.7932).
 - **N-HiTS:** Multi-rate sampling with 3 hierarchical interpolation blocks (0.8049 F1).
 - **WaveNet-GLU:** Dilated causal convolutions ($d = 1, 2, 4, 8$) with Gated Linear Units (31.1% Exact Match).
 - **TFT-Lite:** Softmax Variable Selection Networks (VSN) + Multi-Head Self-Attention (0.8023 F1, 4.018 hrs MAE).
 - **PatchTST:** Sub-series patching ($P = 16, S = 8$) with Channel Independence (0.8041 F1).
 - **Hourly Query Cross-Attention Transformer:** 24 learnable hourly slot queries (0.8056 F1, 90.08% Recall).
5. **Super-Ensemble Stacking:** Weighted probability averaging across top models (**0.8096 F1, 28.9% Exact Match, 4.124 hrs MAE**).

0.6 Summary & Compliance Requirements

- Any new model added to the codebase **MUST** be trained using `train_sota_hourly_models.py` or equivalent standardized pipelines.
- Results **MUST** be saved to JSON/CSV format containing all 8 standardized metrics (`best_hour_level_f1`, `best_hour_level_recall`, `best_hour_level_precision`, `best_hour_level_accuracy`,

best_exact_24h_match_rate, best_mean_absolute_hour_count_error, inference_ms_per_sequence, parameters).

- Reference compiled documentation artifacts:
 - PDF: [docs/standardized_experiment_framework.pdf](standardized_experiment_framework.pdf)
 - Markdown: [docs/standardized_experiment_framework.md](standardized_experiment_framework.md)